

Software Quality Assurance and Management

第三周：验证与确认技术
Week 3: Verification & Validation Techniques

By: Fahad Sabah

计算机学院,
北京工业大学,中国
College of Computer Science,
Beijing University of Technology, China

fahad.sabah@bjut.edu.cn



验证与确认技术

Verification & Validation Techniques

1. Static testing

- ☐ Requirements and design reviews
- ☐ Walkthrough with lab-assistant persona
- ☐ Static analysis (linters and SAST)

2. Dynamic testing

- ☐ Equivalence partitioning (duration and roles)
- ☐ Boundary value analysis (slots and quantity)
- ☐ State transition testing (equipment lifecycle)

3. White-box testing

- ☐ Coverage goals (test plan alignment)
- ☐ Cyclomatic complexity and test sufficiency sketch

4. Exploratory testing

- ☐ Session charter and time box
- ☐ Session notes (exemplar excerpt)
- ☐ Defects logged and session yield
- ☐ Relation to script

1. 静态测试

- ☐ 需求与设计审查
- ☐ 带实验助理角色的演示
- ☐ 静态分析（linters和SAST）

2. 动态测试

- ☐ 等价划分（时长与角色）
- ☐ 边界值分析（槽位和数量）
- ☐ 状态转换测试（设备生命周期）

3. 白盒测试

- ☐ 覆盖目标（测试计划对齐）
- ☐ 环形复杂度与测试充分性草图

4. 探索性测试

- ☐ 会议章程与时间盒
- ☐ 会议记录（示例摘录）
- ☐ 记录的缺陷与会话良率
- ☐ 与文字的关系

验证与确认技术

Verification & Validation Techniques

Verification – built correctly according to specifications

❑ Reviews, static analysis, white-box tests on modules

Validation – right product for the operations

❑ Dynamic tests, UAT, exploratory sessions with real personas

验证—按照规范正确建造

❑ 复习、静态分析、模块白盒测试

验证—适合实验室操作的产品

❑ 动态测试、虚拟身份测试（UAT）、与真实人物的探索性会谈

静态测试—概述

Static Testing – Overview

- ❑ Examine work products **without** running the system
- ❑ Cheaper than late rework

Example – Review yield

- ❑ Design review finds concurrency flaw in $t_{rev} = 2\text{ h}$
- ❑ Fix in production would cost $K_{late} = 7560\text{ CNY}$
- ❑ $Yield = K_{late}/t_{rev} \approx 3780\text{ CNY/h}$ (avoided failure cost for **one** finding)

- ❑ 在不运行系统的情况下检查工作成果
- ❑ 比晚期重做便宜

示例—评审收益率

- ❑ 设计审查发现 $t_{rev} = 2$ 小时的并发缺陷
- ❑ 生产中固定成本为 $K_{late}=7560$ 元人民币
- ❑ $Yield = K_{late}/t_{rev} \approx 3780\text{ CNY/h}$ （一项发现避免失效率）

静态测试—技术

Static Testing – Techniques

1. **Requirements & design reviews** (checklists for security, data rules, conflict policy)
 2. **Walkthrough** of the checkout flow with a lab-assistant persona
 3. **Static analysis** linters + SAST (backend, frontend)
1. **需求与设计审查**（安全检查表、数据规则、冲突政策）
 2. **实验室助理角色的结账流程演示**
 3. **静态分析** linters + SAST（后端、前端）

需求与设计审查

Requirements & Design Reviews

Purpose

Catch risks R1 (overlap), R2 (identity), R4 (QR binding) before sprint closure.

Review checklist clusters for LEBCS design reviews

Theme	Sample review question (LEBCS)
Security	“Check that a student can’t access an admin page using their own login token.” “Make sure the CSV export hides people’s names and emails unless absolutely needed.”
Data rules	“For a given query, ensure the ‘overlap’ condition is written exactly like the acceptance criteria (AC-01).”
Conflict policy	“When two people try to book the same resource at the same time, the database should either lock the row or throw a clear error.”
Audit	“Every time someone checks out, the system must add exactly one new line to the audit log – no edits, no deletions.” (AC-03)?

综述效能—缺陷密度

Review Effectiveness – Defect Density

Metric

$$\delta_{rev} = \frac{n_{find}}{n_{loc}}$$

- n_{find} = defects recorded during review
- n_{loc} = artifact size (pages or KLOC)

Pilot example

12-page API specification $\rightarrow n_{find} = 7$ issues

$$\delta_{rev} = \frac{7}{12} \approx 0.58 \text{ findings/page}$$

Why it matters:

“A high delta_rev (like 0.58) means the review caught many issues early – much more than what would be found after the software is released.”

带实验室助理角色的演示

Walkthrough with Lab-Assistant Persona

Scenario script (checkout desk)

Step 1 – Identity match

“A student shows a QR code. The system scans it and checks that the name matches the one shown from their SSO login.”

Step 2 – State change + audit

“The system moves the item from ‘Reserved’ to ‘CheckedOut’. The assistant looks at the audit log to confirm the timestamp was recorded correctly.”

Step 3 – Late return handling

“If the item is late, a warning banner appears. But the student can still check it in if the rules allow late returns.”

Step 4 – No reservation walk-up

“A student walks up without having reserved anything. The system shows an error code: CHK_NO_RES. This matches requirement AC-02.”

Step 5 – Flaky Wi-Fi resilience

“If the Wi-Fi is unstable during checkout, the system doesn’t break or create two checkouts. The item’s state stays correct (no duplicate checkout).”

场景脚本（结账台）

第一步—身份匹配

“一名学生展示了二维码。系统扫描后确认名字是否与他们的SSO登录显示的匹配。”

第二步—状态变更+审计

“系统会把商品从‘预订’移到‘已借出’。助理查看审计日志，确认时间戳记录正确。”

第三步—晚退货处理

“如果物品迟交，会出现警告横幅。但如果规定允许迟交，学生仍然可以登记。”

第四步—无需预约步行

“一个学生走过来，没有预订任何东西。系统显示错误代码：CHK_NO_RES。这符合AC-02的要求。”

第五步—不稳定的Wi-Fi韧性

“如果结账时Wi-Fi不稳定，系统不会坏掉，也不会生成两个结账。商品状态保持正确（无重复结账）。”

带实验室助理角色的演示

Walkthrough with Lab-Assistant Persona

Clarity score

$$s_{walk} = \frac{b}{a + b} \quad \text{where } (a + b > 0)$$

a = action items opened, b = checklist items passed

Example: $b = 18, a = 4 \Rightarrow s_{walk} \approx 0.82$

- ❑ Meaning: 82% of the steps were clear and passed; 18% needed improvement.
- ❑ Helps quantify documentation quality or usability from a real user's perspective.
- ❑ A higher s_{walk} means fewer action items $\rightarrow$ better clarity.
- ❑ Goal might be $s_{walk} \geq 0.9$ before release.

静态分析—林特与SAST

Static Analysis – Linters & SAST

A **linter** is a static analysis tool that checks for:

- ❑ Coding style (indentation, naming conventions, line length).
- ❑ Potential bugs (unused variables, missing brackets).
- ❑ Code smells (overly complex functions, duplicate code).

Example (JavaScript with ESLint):

Bad code: `function add(a,b){return a+b}`

ESLint (with a recommended style guide) will flag:

- ❑ Missing spaces after commas.
- ❑ Missing space before {.
- ❑ Missing semicolon.

The result helps teams keep the codebase consistent and readable.

SAST (Static Application Security Testing) is a specialized form of static analysis that focuses specifically on security vulnerabilities.

Example (Python with Bandit):

```
password = "super_secret_123" # B105: hardcoded password string
```

A SAST tool will flag this as a high severity issue: hardcoded credentials.

Other common SAST rules detect:

- ❑ SQL injection (unsanitized user input in queries).
- ❑ Use of dangerous functions (eval, pickle).
- ❑ Missing encryption.
- ❑ Cross-site scripting (XSS) patterns in web code.

静态分析—林特与SAST

Static Analysis – Linters & SAST

Key difference:

Both run in the CI pipeline as early checks **before** the code is built and deployed.

Linters	SAST
Focuses on code quality, style, simple bugs	Focuses on security vulnerabilities
Examples: ESLint, Pylint, Checkstyle	Examples: Bandit, Semgrep, SonarQube Security Rules
Often part of a team's coding standards	Often required by compliance (like AC-04, FM-10)

静态分析—林特与SAST

Static Analysis – Linters & SAST

Static analysis toolchain sketch for LEBCS

Layer	Tool class (example)	LEBCS gate
Backend API	Linters + SAST (e.g., ESLint/PyLint + Semgrep/Bandit)	Block merge on high severity; medium reviewed within sprint
Frontend	ESLint + accessibility linter	For checkout components, you cannot use 'any' type. Also, run axe on the main routes that users visit in a normal session.
IaC (Infrastructure as Code) / Docker	hadolint, trivy scan	Any critical vulnerability is not allowed on staging base images unless you get an official waiver.

Static hygiene ratio

$$\eta_{static} = 1 - \frac{N_{sast} - N_{waived}}{N_{sast}} \quad \text{where } (N_{sast} > 0)$$

low risk that you decide not to fix

Example: $N_{sast} = 23, N_{waived} = 5 \rightarrow \eta_{static} \approx 0.217$ (only 22% clean) $\rightarrow$ triggers hardening sprint before UAT.

摘要—静态测试贡献

Summary – Static Testing Contribution

Reviews & walkthroughs catch design/requirement issues early

评测和攻略能及早发现设计/需求问题

Defect density and **clarity score** provide quantifiable feedback

缺陷密度和清晰度评分提供可量化的反馈

SAST gates maintain baseline security & code quality

SAST门保持基线安全和代码质量

All feed into QOTs and acceptance criteria (AC-01–AC-05)

所有输入标准和验收标准 (AC-01–AC-05)

“Verify early, validate often – static techniques are the first line of defense.”

“及早验证，频繁验证—静态技术是第一道防线。”

动态测试概述

Dynamic Testing Overview

- ❑ Execute software with inputs → compare observed vs expected behavior
- ❑ Directly mapped to AC-01–AC-05 (acceptance criteria)
- ❑ **Techniques used:**
 - ❑ Equivalence partitioning (duration, roles)
 - ❑ Boundary value analysis (slot boundaries, quantity limits)
 - ❑ State transition testing (equipment lifecycle)
- ❑ 执行带有输入→比较观察到的行为与预期行为的软件
- ❑ 直接映射到AC-01–AC-05（验收标准）
- ❑ **采用的技术：**
 - ❑ 等价划分（持续时间、角色）
 - ❑ 边界值分析（槽位边界、数量限制）
 - ❑ 状态转换测试（设备生命周期）

等价划分—概念

Equivalence Partitioning – Concept

Myers 划分原则

将输入域划分为不相交类→程序对类中的所有值都一视同仁。
每个类至少选出一名代表。

Myers partition principle

- ❑ Divide input domain into disjoint classes → program treats all values in a class alike.
- ❑ Select at least one representative per class.

LEBCS pilot policy (illustrative) [LEBCS试点政策（示例）]

- Student role: reservation duration d in whole hours, allowed only if $1 \leq d \leq 72$.
- Lab manager role: $1 \leq d \leq 336$ (two weeks).
- Invalid duration: $d \leq 0$ or non-integer d .
- Roles: student, lab_assistant, lab_manager, admin (authorization matrix AC-04).

学生角色的时长划

Duration Partitions for Student Role

Define classes

$$C_{\text{inv-low}} = \{d : d \leq 0\}, \quad C_{\text{val}} = \{d : 1 \leq d \leq 72\}, \quad C_{\text{inv-high}} = \{d : d > 72\}.$$

Representatives

$d \in \{0, 24, 73\} \rightarrow$ three duration tests for student.

Role-duration matrix

Roles: 4(Student, Lab Assistant, Lab manager, Admin) , duration

classes: 3 (valid/invalid bands)

Cross-product = $4 \times 3 = 12$ cells $\rightarrow$ one representative per cell yields 12 cases before boundary refinement.

最小等价套件

Minimal Equivalence Suite

If k independent input dimensions with p_i partitions each:

Weak EP [Equivalence Partition] (one-value-per-partition)

$$N_{EP}^{(\min)} = \sum_{i=1}^k p_i$$

Example – LEBCS

Duration $p_1=3$, Role $p_2=4$, tested independently

$$N_{EP}^{(\min)} = 3 + 4 = 7$$

For full cross-product coverage (role $\times$ duration band)

$$\prod_i p_i = 12$$

示例等价测试案例

Sample Equivalence Test Cases

Table 9: Sample equivalence-class representatives for LEBCS reservation rules.

TC	Role	d (h)	Expected outcome
EP-01	student	0	HTTP 400; no reservation row
EP-02	student	24	HTTP 201; confirmed reservation
EP-03	student	73	HTTP 400; policy message RES_DUR_MAX
EP-04	lab_manager	168	HTTP 201 (within 336 h cap)
EP-05	student	—	POST /admin/equipment → HTTP 403 (AC-04)

One representative per partition → validates role–duration coupling.

HTTP 403
Forbidden error

边界价值分析—概念

Boundary Value Analysis – Concept

Why? Errors cluster on boundaries between partitions.

For a closed interval $[a, b]$ on ordered input x

Two-value BVA: $\{a, b\}$

Three-value BVA:

$$\{a - 1, a, a + 1, b - 1, b, b + 1\}$$

Duration boundaries (student, $a = 1, b = 72$)

Three-value BVA $\rightarrow \{0, 1, 2, 71, 72, 73\}$ – six cases (off-by-one at $d = 72$ and $d = 1$ catch common defects).

developers might accidentally use ≤ 72 instead of < 72

午夜时段边界

Midnight Slot Boundary

Reservation interval $[t_{start}, t_{end})$ in local lab time.

Crosses midnight if

$$date(t_{start}) < date(t_{end})$$

Example

Existing $R1 = [2026 - 05 - 18 \ 22:00, 2026 - 05 - 19 \ 02:00)$

Candidate $R2 = [2026 - 05 - 19 \ 01:00, 03:00) \rightarrow overlaps \ R1 \ (01:00 < 02:00) \rightarrow expect \ HTTP \ 409 \ (AC-01)$

HTTP 409 Conflict

Non-overlapping $R3 = [2026 - 05 - 19 \ 02:00, 04:00)$ borders at 02:00 (half-open) $\rightarrow$ success if policy allows adjacency.

期限边界与数量限制

Term Boundary & Quantity Limits

Term boundary

Invalid cross-term: $R = [2026-06-30 \ 20:00, 2026-07-01 \ 08:00)$

T_1 ends 2026-06-30 23:59, T_2 begins 2026-07-01 00:00

BVA picks last minute inside T_1 and first minute inside T_2 .

Quantity limits

equipment quantity q , cap $q_{max} = 5$

Valid $q \in \{1, 2, \dots, 5\}$

BVA on q : $\{0, 1, 2, 5, 6\}$ (invalid 0 and 6)

For $q = 3$: three concurrent non-overlapping reservations succeed; a fourth overlapping fails.

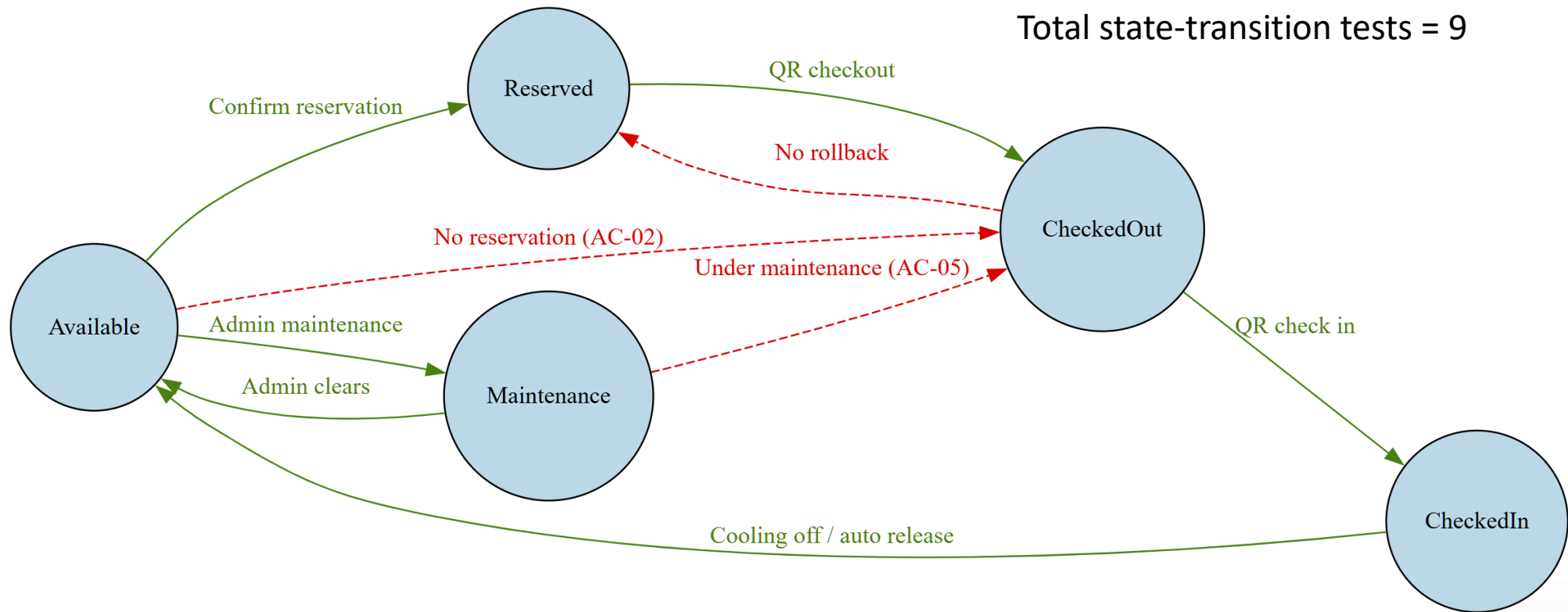
状态转换测试—设备生命周期

State Transition Testing – Equipment Lifecycle

States

$S = \{Available, Reserved, CheckedOut, CheckedIn, Maintenance\}$

Green solid arrows → valid transitions (6 tests)
Red dashed arrows → invalid transitions (3 tests)
Total state-transition tests = 9



Week 3: Verification & Validation Techniques

第三周：验证与确认技术

状态转换测试—设备生命周期

State Transition Testing – Equipment Lifecycle

Valid transitions (MVP) – each row becomes an integration test (QOT-01, QOT-02)

From	To	Trigger	AC / risk
Available	Reserved	Confirm reservation	AC-01
Reserved	CheckedOut	QR checkout	AC-02, AC-03
CheckedOut	CheckedIn	QR check-in	AC-03
CheckedIn	Available	Cooling-off / auto-release	–
Available	Maintenance	Admin maintenance window	AC-05
Maintenance	Available	Admin clears maintenance	AC-05

无效的转移 (示例)

Invalid Transitions (sample)

- ❑ (Available, CheckedOut) without reservation → must fail (AC-02)
- ❑ (Maintenance, CheckedOut) → must fail (AC-05)
- ❑ (CheckedOut, Reserved) → must fail (no rollback without admin tool)

Transition coverage metric

$|S|=5$,
 $|T|=6$ valid edges

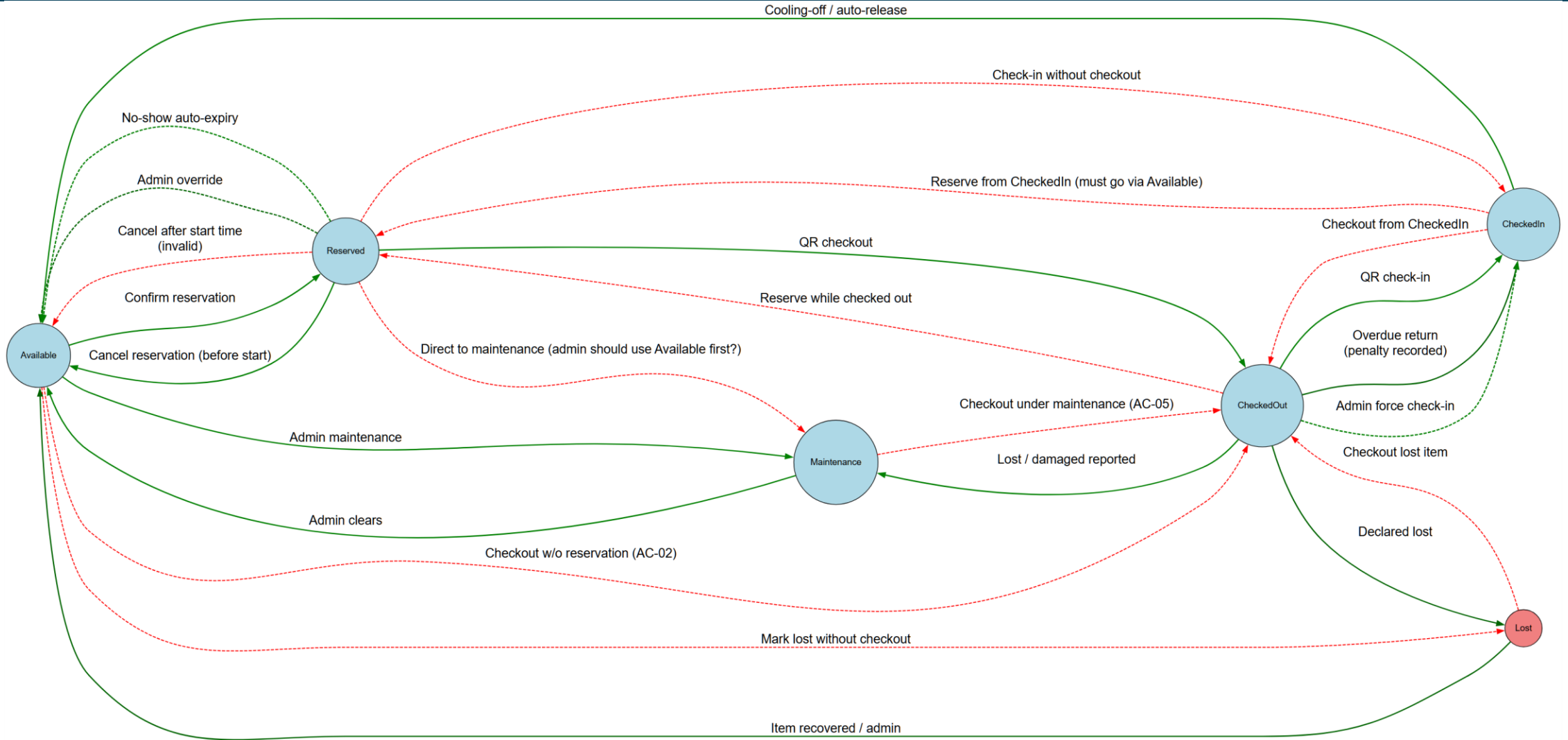
- ❑ (可用, 已借出) 无预订→必须失败 (AC-02)
- ❑ (维护, 检查完毕) →必须故障 (AC-05)
- ❑ (已退房, 保留) →必须失败 (没有管理员工具不可回滚)

过渡覆盖度

$|S|=5$,
 $|T|=6$ 条有效边

状态转换测试

State Transition Testing



并发笔记 (风险R1)

Concurrency Note (Risk R1)

Two parallel checkout attempts from Reserved state:

- ☐ Exactly one CheckedOut transition succeeds
- ☐ Other request receives rejection response
- ☐ Dynamic complement to white-box tests on conflict resolver

从预留状态中出现的两次平行结账尝试:

- ☐ 只有一次 CheckedOut 转换成功
- ☐ 其他请求收到拒绝回复
- ☐ 冲突解决器白盒测试的动态补充

摘要—动态测试

Summary – Dynamic Testing

- ❑ 等价划分涵盖角色/持续时间类别（12+情况）
- ❑ 边界价值分析针对午夜、期限、数量限制（6+个案例）
- ❑ 状态转换测试验证生命周期边缘（9个基础测试）
- ❑ 所有目标直接映射到AC-01–AC-05，并输入测试目标QOT-01–QOT-09
- ❑ “测试边界，穿越州际，明智划分。”

- ❑ Equivalence partitioning covers role/duration classes (12+ cases)
- ❑ Boundary value analysis targets midnight, term, quantity limits (6+ cases)
- ❑ State transition testing validates lifecycle edges (9 basic tests)
- ❑ All map directly to AC-01–AC-05 and feed test objectives QOT-01–QOT-09
- ❑ "Test boundaries, traverse states, partition wisely."

白箱测试

White-box Testing

LEBCS – Conflict Resolver Structural Coverage

White-box Testing – Focus

Uses knowledge of control flow inside the reservation conflict resolver

The resolver decides:

- ☐ Whether interval $[t_a, t_b)$ overlaps existing confirmed reservations for equipment e when quantity $q = 1$
- ☐ Whether capacity remains when $q > 1$ (AC-01, FM-01)

LEBCS – 冲突解决者结构性保障

白盒测试一焦点

利用预订冲突解决器内的控制流程知识

解析器决定:

- ☐ 当数量 $q=1$ 时, 区间 $[t_a, t_b)$ 是否与设备 e 已有确认预留重叠
- ☐ 容量是否在 $q>1$ (AC-01, FM-01) 时保持不

覆盖目标（测试计划对齐）

Coverage Goals (Test Plan Alignment)

Let:

N_{stmt} =executable statements in module

N_{br} =decision outcomes (branches)

n_{stmt}, n_{br} = covered after tests

Statement coverage C_0 and branch coverage C_1 :

$$C_0 = \frac{n_{stmt}}{N_{stmt}}, C_1 = \frac{n_{br}}{N_{br}}$$

Pilot goals (release candidate)

- ☐ $C_0 \geq 0.90$
- ☐ $C_1 \geq 0.85$ on the conflict resolver package
- ☐ Measured by language coverage tool in CI

设:

N_{stmt} =模块中的可执行语句

N_{br} =决策结果（分支）

n_{stmt}, n_{br} =测试后覆盖

声明覆盖 C_0 and 分支保险覆盖 C_1 :

$$C_0 = \frac{n_{stmt}}{N_{stmt}}, C_1 = \frac{n_{br}}{N_{br}}$$

试点目标（发布候选）

- ☐ $C_0 \geq 0.90$
- ☐ $C_1 \geq 0.85$ 关于冲突解决器包
- ☐ 通过CI中的语言覆盖工具进行测量

示例

Worked Example

Suppose:

$$N_{stmt} = 40, N_{br} = 24$$

After unit + integration tests:

$$n_{stmt} = 37, n_{br} = 21$$

Then:

$$C0 = 37/40 = 0.925 (met, \geq 0.90)$$

$$C1 = 21/24 = 0.875 (met, \geq 0.85)$$

If one branch remains uncovered (e.g., "overlap at exact endpoint" path) → add test with bordering intervals from BVA (Midnight Slot Boundary example).

假设:

$$N_{stmt} = 40, N_{br} = 24$$

单元+积分测试后:

$$n_{stmt} = 37, n_{br} = 21$$

然后:

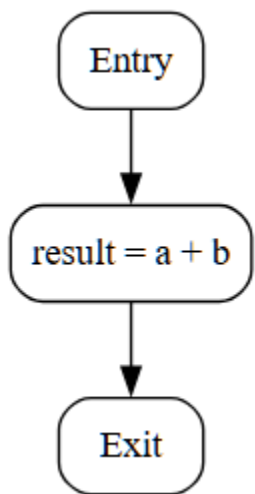
$$C0 = 37/40 = 0.925 (met, \geq 0.90)$$

$$C1 = 21/24 = 0.875 (met, \geq 0.85)$$

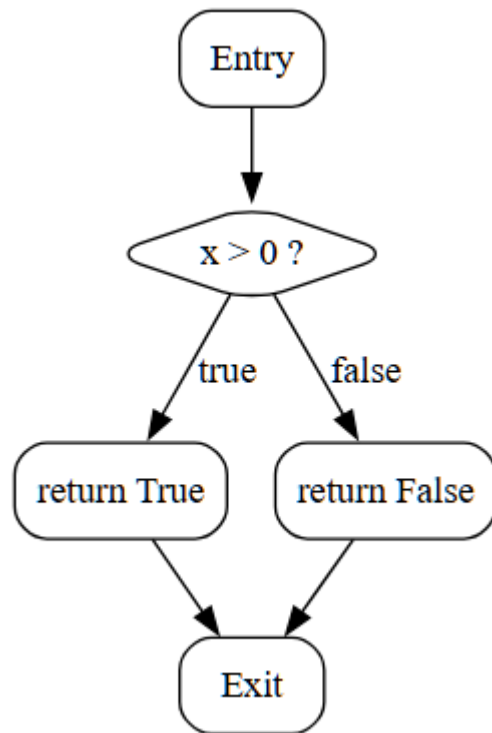
如果某条分支仍未覆盖（例如“精确端点重叠”路径），→添加与BVA相邻区间的检验（午夜时隙边界示例）。

环形复杂性

Cyclomatic Complexity



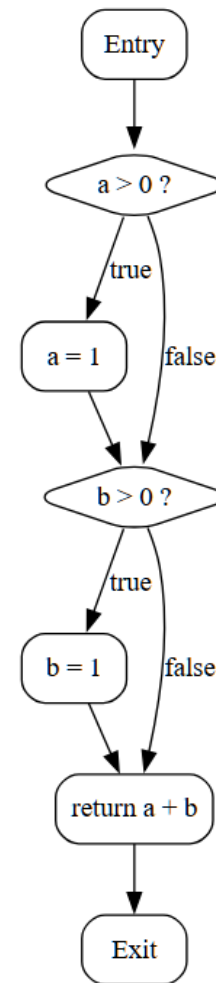
$n=2, e=1 \rightarrow V=1-2+2=1$
Minimum tests: 1



$n=5, e=5 \rightarrow V=5-5+2=2$
Minimum tests: 2 (e.g., $x=1$ and $x=-1$)

$n=7, e=8 \rightarrow V=8-7+2=3$
Minimum tests: 3 independent paths:

- ☐ $(a>0, b>0)$
- ☐ $(a>0, b\leq 0)$
- ☐ $(a\leq 0, b>0)$



环形复杂性

Cyclomatic Complexity

For a control-flow graph with e edges, n nodes, p connected components:

$$V(G) = e - n + 2p$$

(McCabe's cyclomatic complexity)

Classical guideline: unit tests should exercise at least $V(G)$ linearly independent paths.

$V(G)$ = number of decision points (conditions, loops, switches) + 1.

对于一个控制流图，满足 e 条边， n 个节点， p 个连通分量：

$$V(G) = e - n + 2p$$

(麦凯布的环形复杂性)

经典指导原则：单元测试应至少运行 $V(G)$ 线性独立路径。

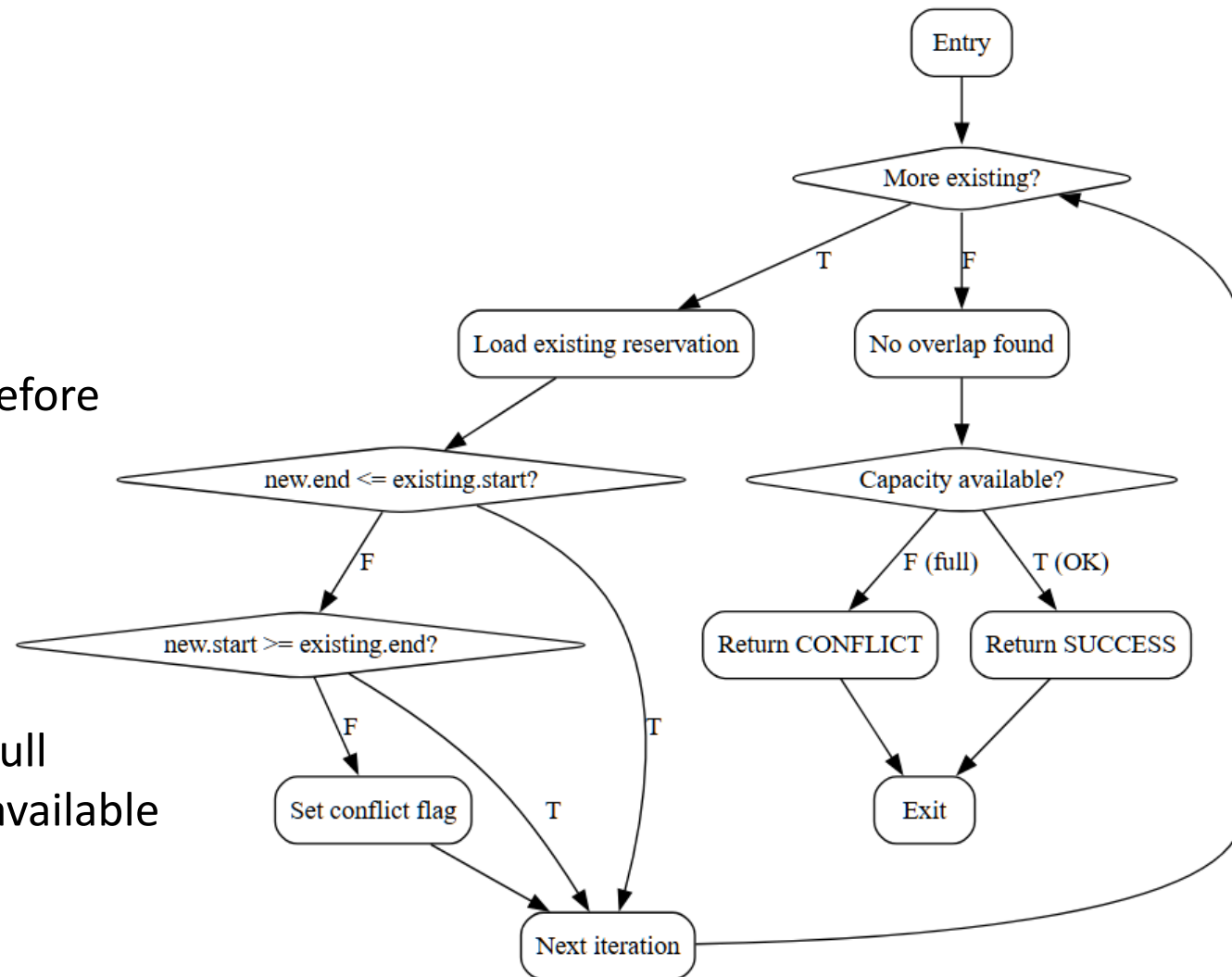
插图

Illustration

$$n = 12, e = 15$$
$$V(G) = 15 - 12 + 2 = 5$$

Plan five path-focused unit tests (before exception paths):

1. No overlap
2. Overlap partial
3. Overlap full
4. Quantity $q > 1$ – capacity full
5. Quantity $q > 1$ – capacity available



插图

Illustration

$$n = 12, e = 15$$
$$V(G) = 15 - 12 + 2 = 5$$

Plan five path-focused unit tests (before exception paths):

1. No overlap
2. Overlap partial
3. Overlap full
4. Quantity $q > 1$ – capacity full
5. Quantity $q > 1$ – capacity available

$$n = 12, e = 15$$
$$V(G) = 15 - 12 + 2 = 5$$

规划五个以路径为中心的单元测试（例外路径之前）：

1. 没有重叠
2. 重叠部分
3. 重叠全场
4. 数量 $q>1$ —满载能力
5. 数量 $q>1$ —可用容量

结构测试设计—重叠谓词

Structural Test Design – Overlap Predicate

Half-open intervals $[t_1, t_2)$ and $[t_3, t_4)$:

$$\text{Overlap}(t_1, t_2, t_3, t_4) \equiv (t_1 < t_4) \wedge (t_3 < t_2)$$

Each conjunct is a boolean decision with two branches.

Full branch coverage requires tests where each clause is true and false independently $\rightarrow$ four combinations (minus infeasible).

半开音程 $[t_1, t_2)$ 和 $[t_3, t_4)$:

$$\text{Overlap}(t_1, t_2, t_3, t_4) \equiv (t_1 < t_4) \wedge (t_3 < t_2)$$

每个合取是一个布尔判定，有两个分支。

全分支覆盖需要测试，使每个子句在四个组合 $\rightarrow$ 独立为真和假（减去不可行）。

LEBCS 对谓词的扩展

LEBCS Extensions to the Predicate

Guard for status = confirmed

$q=1 \rightarrow$ strict overlap rule

$q>1 \rightarrow$ capacity counter (count overlapping confirmed reservations)

These additions double the decision points $\rightarrow$ motivates $C1 \geq 0.85$ (not just statement coverage).

警卫身份 = 确认

$q=1 \rightarrow$ 严格重叠规则

$q>1 \rightarrow$ 容量计数器（确认预订数量重叠）

这些新增内容使决策点 $\rightarrow$ 动机加倍
 $C1 \geq 0.85$ （不仅仅是账单覆盖率）。

摘要—白盒测试

Summary – White-box Testing

- ❑ Statement coverage ($C_0 \geq 0.90$) ensures most lines execute
- ❑ Branch coverage ($C_1 \geq 0.85$) targets decision logic
- ❑ Cyclomatic complexity ($V(G)$) guides minimum number of path tests
- ❑ Overlap predicate + capacity rules drive structural test design
- ❑ "Know the paths, cover the branches, catch the conflicts."
- ❑ 语句覆盖率 ($C_0 \geq 0.90$) 确保大多数行执行
- ❑ 分支覆盖率 ($C_1 \geq 0.85$) 针对决策逻辑
- ❑ 环状复杂度 ($V(G)$) 指导路径测试的最少次数
- ❑ 重叠谓词+容量规则驱动结构测试设计
- ❑ “熟悉路径，遮蔽树枝，捕捉冲突。”

探索性测试

Exploratory Testing

LEBCS – Session-Based Discovery

Combines learning, test design, and execution in real time

Complements scripted EP/BVA/STM

Probes human–system interaction under realistic desk pressure (risk R5, WP 8.2)

Representative charter: "Break checkout under flaky network"

LEBCS – 基于会话的发现

实时结合学习、测试设计和执行

补充剧本EP/BVA/STM

在现实桌面压力下探究人与系统交互
(风险R5, WP 8.2)

代表特许：“在不稳定网络下破解结账”

会议章程与时间盒

Session Charter and Time Box

Title: Break checkout under flaky network

Mission: Discover failures in checkout/check-in under campus Wi-Fi profile P1 (latency, packet loss, brief disconnects)

Scope:

In: QR scan, confirm dialog, double-tap, back-button during spinner
Out: IdP certificate expiry (covered elsewhere)

Time box: 90 minutes, two testers (QA + lab assistant persona)

Oracles:

- ☐ AC-02 – no checkout without reservation
- ☐ AC-03 – exactly one audit row per successful checkout
- ☐ No duplicate CheckedOut state (FM-09)

标题: 不稳定网络下的破解结账

任务: 发现校园Wi-Fi配置文件P1下的结账/签到失败（延迟、丢包、短暂断线）

范围:

在：二维码扫描、确认对话、双击、旋转时返回键

退出: IdP证书过期（其他地方有报道）

时间框：90分钟，两名测试员（QA + 实验室助理角色）

神谕者:

- ☐ AC-02 一无预订不退房
- ☐ AC-03—每次成功结账仅需一行审计
- ☐ 无重复的CheckedOut状态（FM-09）

会议记录（范例摘录）

Session Notes (Exemplar Excerpt)

Time	Action	Observation
0:00	Enable 3G throttling + 5% loss	Login succeeds; catalog slow but usable
0:12	Scan QR; tap Confirm twice quickly	One checkout; second HTTP 409 idempotent – Pass
0:28	Disconnect Wi-Fi at spinner; reconnect	UI stuck; refresh shows CheckedOut – Fail (DEF-142)
0:41	Retry checkout same reservation	Confusing error CHK_ALREADY – usability note
1:05	Check-in under loss	Audit row present; UI toast delayed 8s – Pass with UX note

记录的缺陷与会话性能

Defects Logged and Session Yield

Let

$d_{sev1}, d_{sev2}, d_{sev3}$ be defects filed by severity.

Session yield (defects per hour):

$$y_{session} = \frac{d_{sev1} + d_{sev2} + d_{sev3}}{t_{session}/60}$$

($t_{session}$ in minutes)

Example:

$$d_{sev1} = 0, d_{sev2} = 1, d_{sev3} = 2, d_{session} = 90$$

$$y_{session} = 3/1.5 = 2 \text{ defects/hour}$$

设

$d_{sev1}, d_{sev2}, d_{sev3}$ 为按严重程度提交的缺陷。

会话产量（每小时缺陷数）：

$$y_{session} = \frac{d_{sev1} + d_{sev2} + d_{sev3}}{t_{session}/60}$$

($t_{session}$ in minutes)

示例：

$$d_{sev1} = 0, d_{sev2} = 1, d_{sev3} = 2, d_{session} = 90$$

$$y_{session} = 3/1.5 = 2 \text{ defects/hour}$$

样本发现

Sample Findings

- ❑ DEF-142 (Sev-2): After connectivity loss during checkout spinner, client does not reconcile server state; user may attempt duplicate actions.
- ❑ → Linked to FM-09, R5. Retest on staging build 1.0.0-rc3.
- ❑ DEF-143 (Sev-3): Error copy for CHK_ALREADY does not suggest check-in path.
- ❑ DEF-144 (Sev-3): Toast delay >5s under loss; monitor only if median task time fails pilot target.
- ❑ DEF-142 (Sev-2) : 在结账旋转器中断连接后, 客户端无法对服务器状态进行对账; 用户可以尝试重复操作。
- ❑ → 关联到FM-09, R5。在分阶段构建1.0.0-rc3上重新测试。
- ❑ DEF-143 (Sev-3) : CHK_ALREADY的错误副本不建议签入路径。
- ❑ DEF-144 (Sev-3) : 吐司延迟>损失5秒; 仅在任务时间中位数未达标时进行监控。

与脚本组曲的关系

Relation to Scripted Suites

Exploratory sessions do not replace EP/BVA/STM – they feed them.

探索性课程并不能取代EP/BVA/STM—它们是为他们提供能量。

DEF-142 motivates a new scripted case: "disconnect at confirm" in API/UI regression pack (WP 6.2)

DEF-142 引发了一个新的脚本化案例：“在API/UI 回归包（WP 6.2）中确认时断开连接”

Feedback ratio:

$$r_{feed} = \frac{n_{script_added}}{d_{total}}$$

反馈比率：

$$r_{feed} = \frac{n_{script_added}}{d_{total}}$$

Example: $1/3 \approx 0.33$ of session defects became formal tests → healthy hardening loop.

示例：1/3≈0.33的会话缺陷成为了→健康硬化环的正式测试。

总结

Summary



项目讨论

Project Discussion

❑ Project Type 1

1. SQAM_Portfolio
2. Need to add group members/authors details in latex template/project provided
3. Add/update the relevant sections/content discussed in the class

❑ Project Type 2

1. Research/Survey Paper
2. Need to add group members/authors details in latex template/project provided
3. Set the title of the project
4. Add search query to the project.
5. Install the tools (Rstudio, VoSViewer etc.)
6. Perform analyses as discussed.